DATA ANALYSIS AND INTERPRETATION / GROUP 13

Does Height Tell the Whole Story?

A height–weight investigation among Indian Summer Olympic athletes

Question 2 • Sports and Biometrics

The question we set out to answer. *Does the height–weight relationship differ across sports after accounting for sex?*

Imagine two athletes who are both 175 cm tall. One competes in distance running, the other in a power-oriented event. Would a single height-based formula predict their weights equally well? Intuition suggests that sporting demands might matter. But intuition is not evidence—and a visible difference on a scatterplot does not automatically mean a difference in the underlying relationships.

We approached this question in stages: start with the simplest possible explanation, look for what it misses, then ask whether the added complexity is justified.

1. First, is there even a relationship to explain?

Across 732 athlete-year observations, height and weight have a Pearson correlation of $r = 0.698$. The positive association is hard to miss: taller athletes generally weigh more. Yet the scatter is far from a narrow line. At similar heights, observed weights can differ substantially.

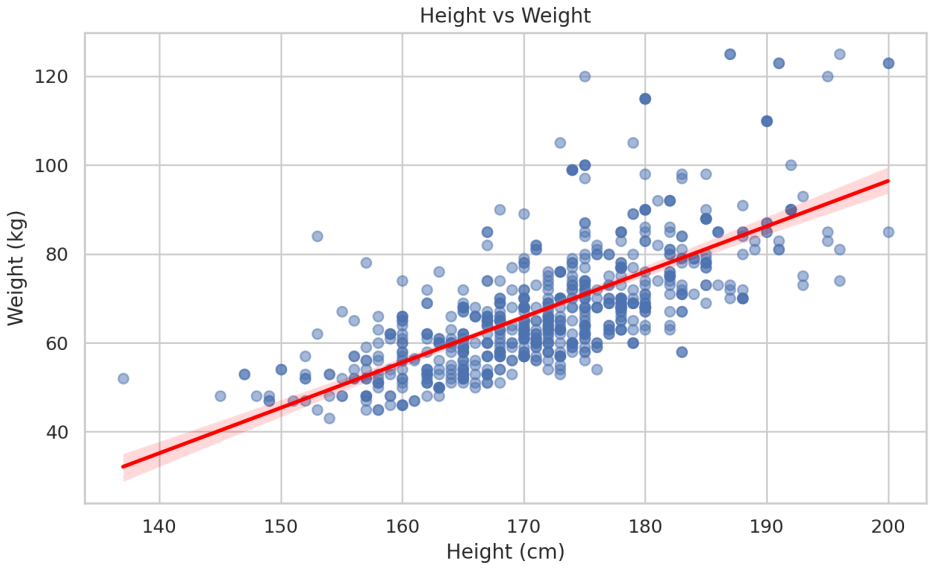


Figure 1: Height and weight follow a clear upward trend, but athletes of similar height do not all have similar weights.

That immediately raises our first challenge: *how much of weight can height explain on its own?*

2. What happens if we insist on one line for everyone?

We began with a simple linear regression, where weight W (kg) is predicted from height H (cm):

$$\widehat{W} = -107.782 + 1.021H.$$

The slope suggests an increase of about 1.02 kg in predicted weight for each additional centimetre of height in the pooled data. Its $R^2 = 0.487$, meaning that height alone accounts for about 48.7% of the observed weight variation in this sample. The negative intercept is merely part of the fitted line: extrapolating it to an athlete of zero height has no physical meaning.

This is a useful starting point, not an answer to our central question. A single line may capture the overall trend while systematically missing particular types of athletes.

3. Who does the single line get wrong?

To find out, we examined *residuals*: actual weight minus predicted weight. A positive residual means the athlete is heavier than the baseline prediction; a negative one means the model predicts too much weight.

The errors were not distributed identically across archetypes. Endurance athletes had a mean residual of -7.27 kg, whereas Combat and Precision had mean residuals of $+3.36$ and $+3.87$ kg, respectively. In other words, the same pooled line tended to overpredict Endurance weights and underpredict weights in Combat and Precision.

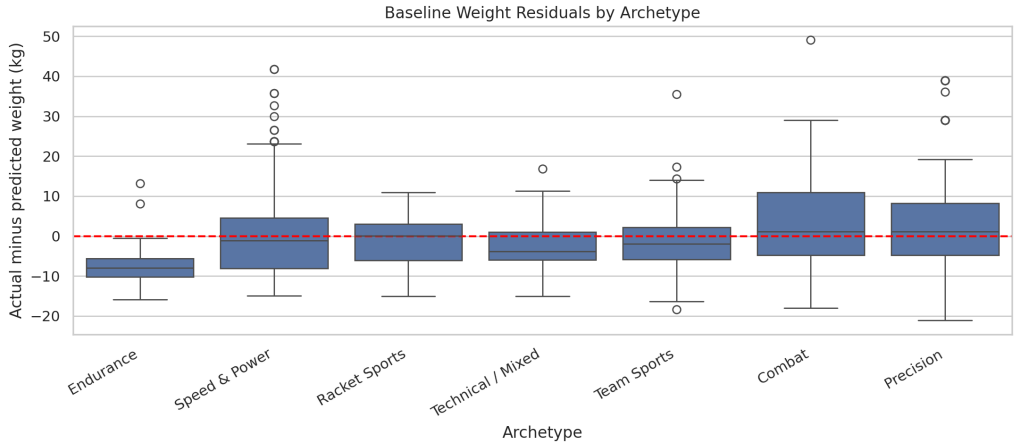


Figure 2: Baseline residuals grouped by functional archetype. The groups show different typical departures from a single pooled regression line.

That pattern is the turning point of the analysis. *If the errors know which archetype an athlete belongs to, should the regression know too?*

4. Could the relationship itself be different?

We next fitted a multiple linear regression using height, sex, and functional archetype. Crucially, we also allowed the *slope of height* to change by archetype, not just the predicted weight at a fixed height. The model can be written as

$$W_i = \beta_0 + \beta_H H_{c,i} + \sum_{j=1}^{6} \gamma_j A_{ij} + \theta F_i + \sum_{j=1}^{6} \delta_j (H_{c,i} A_{ij}) + \varepsilon_i.$$

Here H_c is height centred at the modelling-sample mean, A_j indicate sports archetypes, and F indicates female sex. Endurance and male are the reference categories. The interaction coefficients δ_j

are the key quantities: each represents how much an archetype's height slope differs from Endurance's slope.

The extended model uses 727 athlete-year observations; five observations assigned to multiple archetypes were excluded rather than arbitrarily placed in a single group. The data contain repeated Olympic-year appearances for some athletes, so these are athlete-year records, not 727 unique people.

Table 1: The estimated height slope is not the same in every archetype.

Archetype	Estimated slope (kg per cm)
Endurance	0.886
Combat	1.218
Precision	1.017
Racket Sports	0.707
Speed & Power	1.261
Team Sports	0.613
Technical / Mixed	0.701

Consider the contrast between Speed & Power (about 1.26 kg/cm) and Team Sports (about 0.61 kg/cm). These are *estimated associations*, not physiological laws or causal effects. Also, sport-specific variation within each broad archetype remains hidden by the grouping.

On the common 727-record sample, the baseline model has $R^2 \approx 0.487$ and the extended model has $R^2 \approx 0.576$. Including sex, archetype, and differing slopes describes more of the observed variation. In the fitted model, the female indicator has a coefficient of about -3.83 kg at fixed height and archetype; because the model contains no sex interactions, it assumes this conditional difference is shared across archetypes.

But an important doubt remains: *are different estimated slopes sufficient to reject the idea of one common slope?*

5. Could these slope differences be chance variation?

This is the formal hypothesis test directly tied to our Checkpoint 1 question. We compare a reduced regression with a common height slope to the full model with six height-by-archetype interactions:

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_6 = 0,$$
$$H_1 : \text{at least one interaction coefficient is nonzero.}$$

A nested-model partial F-test, fitted on the same 727 observations, yields

$$\boxed{F(6, 712) = 9.4019, \qquad p \approx 6.01 \times 10^{-10}.}$$

At the 5% significance level, we reject the common-slope null hypothesis. **The data provide strong evidence that the height–weight association differs across functional archetypes after accounting for sex.** This does not identify every pair of groups that differs, and it does not show that playing a particular sport causes a particular body weight.

This is an ordinary OLS F-test. Because some athletes contribute multiple years, within-athlete dependence may affect the usual reference distribution; we therefore treat the result as evidence under the OLS assumptions, not as an assumption-free guarantee.

6. Before trusting the fitted lines, what do their errors look like?

A fitted regression can describe a trend while making imperfect assumptions about individual errors. The residual-versus-fitted plot reveals their spread; Q–Q plots ask whether the residual distribution resembles a normal distribution, especially at the extremes.

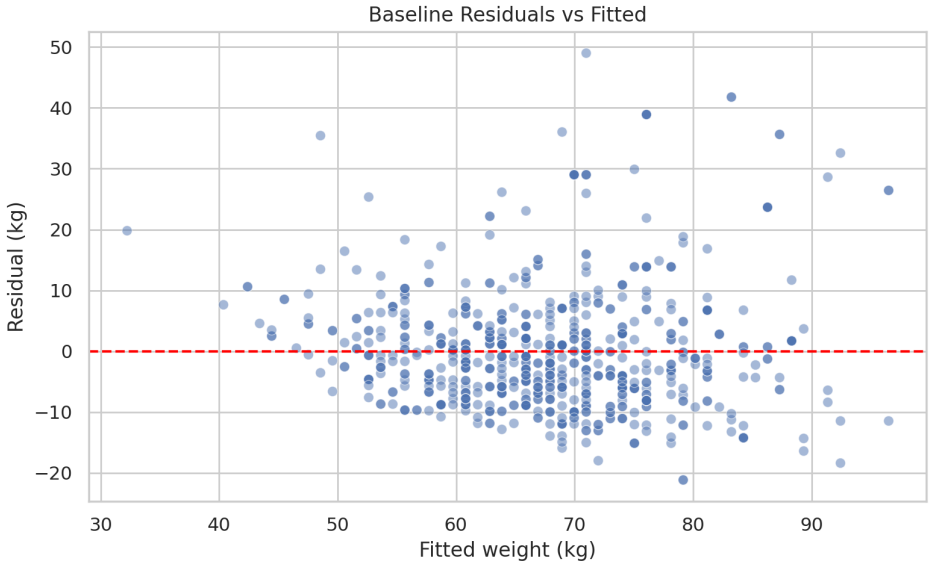


Figure 3: Residuals versus fitted values for the height-only regression.

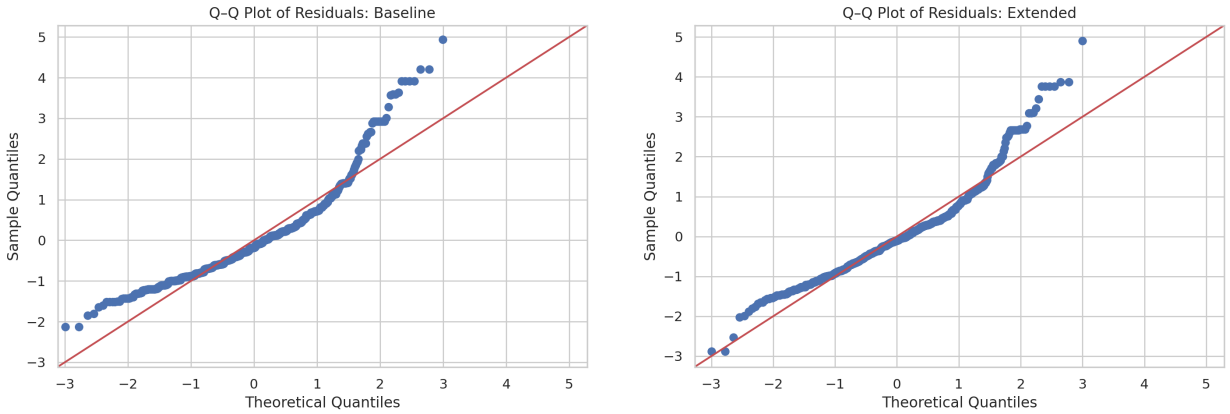


Figure 4: Q–Q plots for baseline (left) and extended (right) residuals. Departures in the tails caution against treating normal-error assumptions as exact.

The Q–Q plots are exploratory checks rather than additional hypothesis tests. Their departures from the reference line, especially near the tails, remind us that a better average fit does not eliminate atypical athletes or make uncertainty disappear.

7. What if we predict athletes the model has never seen?

We next held out approximately 20% of *athletes*, not merely 20% of rows. This matters because the same athlete may appear in more than one Olympic year: putting that athlete in both training and testing would make the evaluation less independent.

The split contains **580 training athlete-year observations from 416 athletes** and **147 test observations from 105 athletes**. The models were fitted on the training portion; for the extended regression, height was centred using the training-set mean only. The held-out records let us inspect

predictions for athletes absent from model fitting, without introducing aggregate performance metrics.

Now comes a more human question: *if a model predicts 70 kg, should we expect the athlete to weigh almost exactly 70 kg?*

8. How uncertain is one athlete's predicted weight?

To avoid presenting a point prediction as a certainty, we calculated conventional **95% individual prediction intervals** using the model fitted on the training sample. Such intervals are wider than confidence intervals for the mean response because they must accommodate variation among individual athletes, not just uncertainty in the regression line.

Table 2: Illustrative held-out predictions from the extended regression (kg).

Athlete ID	Olympic year	Actual	Predicted	Individual 95% interval
512	1992	53	53.23	[34.28, 72.17]
512	1996	53	61.43	[42.83, 80.04]
6469	2016	52	59.26	[40.50, 78.02]
8458	1972	75	68.00	[48.78, 87.23]
11197	1976	68	67.79	[49.27, 86.32]

For the 1992 record of athlete 512, the predicted weight is 53.23 kg, but the conventional interval stretches from 34.28 to 72.17 kg. The broad range is not a failure to report a precise enough decimal; it reflects the substantial uncertainty involved in estimating one person's weight from a small set of characteristics. The intervals rely on the usual OLS error assumptions and are not adjusted for dependence between repeated athlete-year observations or unequal residual variation.

So, does height tell the whole story?

No—and this is the main finding. Height has a meaningful positive association with weight ($r = 0.698$), but a single pooled regression misses systematic differences among sports archetypes. Allowing the height slope to vary across archetypes improves in-sample fit (R^2 from approximately 0.487 to 0.576 on a common sample), and the interaction F-test rejects a shared slope under ordinary OLS assumptions.

Yet the story also has limits. Our data describe recorded Indian Summer Olympic athlete-year observations, not every athlete; categories combine sports with different physical demands; some categories are small; and repeated observations and non-normal residual tails complicate formal inference and prediction intervals. The evidence supports a more nuanced *association* between height and weight—not a claim that sport or sex causes a particular body weight.

Takeaway: The question is not simply whether taller athletes weigh more. It is whether *the same centimetre of height means the same thing in every sporting context.* In these data, it does not.